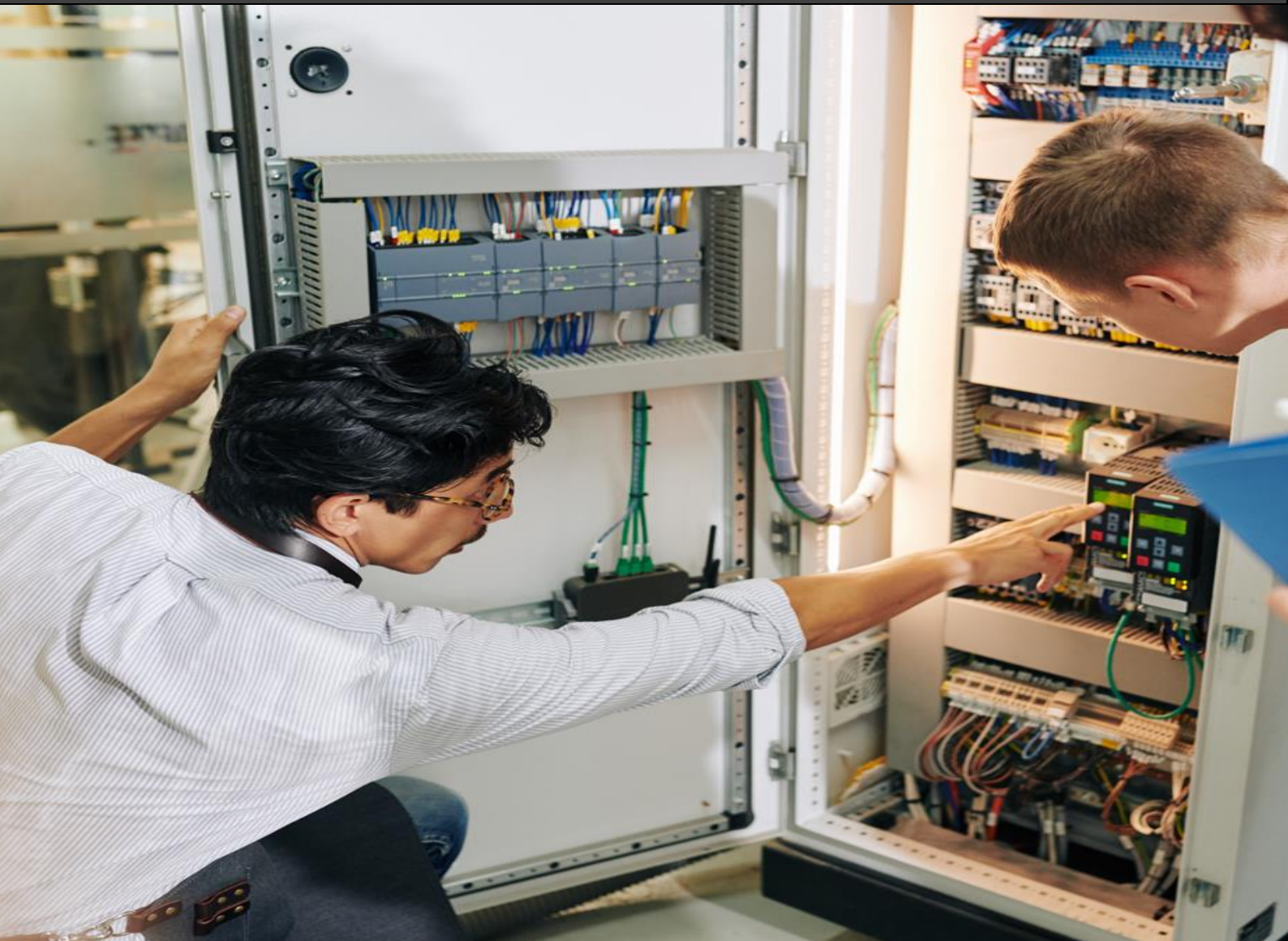


PLC Fundamentals for Maintenance Engineers - Answers



MATRIX

CP2388

www.matrixtsl.com

Copyright © 2025 Matrix Technology Solutions Limited

Introduction 2

Answers 3

Worksheet 2 – Complex PLC Control Systems 5

Worksheet 3 – HMIs 6

Worksheet 4 – Emergency Stops 7

Worksheet 5 – Status LED 8

Worksheet 6 – Normally Open vs Normally Closed 9

Worksheet 7 – Proximity Switch 10

Worksheet 8 – Potentiometer 11

Worksheet 9 – Temperature Sensor 12

Worksheet 10 – Digital Outputs 13

Worksheet 11 – Motor PWM 14

Worksheet 12 – Motor Relay 15

Version Control 17

Introduction

INTRODUCTION

This document highlights all the answers to the questions posed in the student handout for the worksheets curriculum “CP2388 - PLC Fundamentals for Maintenance Engineers”.



ANSWERS

Answers

Worksheet 1 – Simple PLC Control Systems

Questions:

1. What type of control does this system demonstrate?
2. What happens when you press and hold input I0?
3. What do the PLC's status LEDs show during operation?
4. Where might a system like this be used in real industrial settings?

Answers:

1. It demonstrates simple on/off control using push buttons and indicator lights.
2. The red light turns on while the button is pressed, then turns off when released.
3. They show which inputs and outputs are active, based on the PLC logic.
4. In factories to control devices like lights, fans, or alarms.

Answers

Worksheet 2 – Complex PLC Control Systems

Questions:

1. What does a complex PLC system control?
2. What happens when you press input I0 in this worksheet?
3. What does it mean when the PLC latches a signal?
4. How can the process be stopped once it starts?
5. Why is understanding PLC logic important for fault finding?

Answers:

1. It controls step-by-step processes using inputs, outputs, and logic, like in real factory systems.
2. The outputs turn on one by one in a sequence, even after you release the button.
3. The PLC stores the input signal in memory, so the process keeps running without holding the button.
4. By pressing the red stop button (I2) or the emergency stop.
5. Because most real systems are complex, and you need to know how the program works to fix problems.

Answers

Worksheet 3 – HMIs

Questions:

1. What is the purpose of an HMI in an industrial system?
2. What does this worksheet's HMI screen represent?
3. What happens when you press the Mix button?
4. Why are some HMI controls greyed out during mixing?
5. How can maintenance engineers use HMIs during fault finding?

Answers:

1. An HMI lets operators view system status and send commands to the PLC.
2. It shows a batch mixing process with ingredient tanks and a final product tank.
3. The three tanks drain into the bottom tank, and controls are locked until mixing finishes.
4. To prevent changes during critical parts of the process, keeping the system safe.
5. They use it to check inputs, outputs, alarms, and test system behaviour.

Answers

Worksheet 4 – Emergency Stops

Questions:

1. What is the purpose of an Emergency Stop (E-Stop)?
2. How is an E-Stop different from a normal stop button?
3. What happens when you press the E-Stop during the mixing process?
4. Why does the E-Stop have two contacts?
5. What steps must be taken to reset the system after pressing the E-Stop?

Answers:

1. To stop the machine immediately in an emergency to keep people safe and protect equipment.
2. An E-Stop is only for emergencies, not for normal stopping or shutdown.
3. The system stops, an alarm appears, and mixing is halted until reset.
4. For safety. If either contact opens, the system stops. This makes the E-Stop more reliable.
5. Twist the E-Stop to release it, then press the blue reset button or use the HMI reset.

Answers

Worksheet 5 – Status LED

Questions:

1. What is the purpose of the status LED in a PLC system?
2. What does a red status LED usually mean?
3. What does a green status LED usually mean?
4. How does the status LED change across Basic, Enhanced, and Live Process modes?
5. Why is it important to understand how the LED is programmed?

Answers:

1. It shows the current state of the system using colours, helping identify faults or normal operation.
2. It often means there is a fault or the emergency stop has been pressed.
3. It usually means the system is healthy and ready or running normally.
4. It may show solid or flashing patterns based on system state, like startup, running, or fault.
5. Because LED behaviour is set in the PLC code and can vary between machines, so reading the manual and understanding the logic is key.

Answers

Worksheet 6 – Normally Open vs Normally Closed

Questions:

1. What is the difference between a normally open (NO) and normally closed (NC) contact?
2. What does green text on the HMI input screen mean?
3. How does Input 2 behave when not pressed?
4. What happens to Input 3 when it is pressed and released?
5. Why is it important to understand NO and NC input states in maintenance?

Answers:

1. An NO contact is off until pressed, while an NC contact is on until pressed.
2. Green means the input is currently on; white means it is off.
3. It stays on by default and turns off when the red button is pressed.
4. It latches on when pressed and stays on until pressed again.
5. So you can correctly read input states and avoid misdiagnosing faults.

Answers

Worksheet 7 – Proximity Switch

Questions:

1. What does a proximity sensor detect?
2. Why are proximity sensors useful in automation systems?
3. What type of object does an inductive sensor detect?
4. What caused the fault in the locking sequence during the exercise?
5. How can engineers test a proximity sensor during troubleshooting?

Answers:

1. It detects the presence of an object without physical contact.
2. They improve safety and reduce wear by detecting position without touching the object.
3. Metal objects only.
4. The system couldn't detect the locking pins because the sensor was not triggered.
5. By placing a metal object in front of the sensor to simulate detection.

Answers

Worksheet 8 – Potentiometer

Questions:

1. What does a potentiometer do in a PLC system?
2. What voltage range does a potentiometer typically output?
3. How does the motor respond as you turn the potentiometer?
4. What does applying an offset do to the input signal?
5. Why is scaling (scalar) important in PLC systems?

Answers:

1. It creates a variable voltage signal used to control things like motor speed or level settings.
2. Between 0 and 10 volts.
3. The motor speed increases as the potentiometer is turned to the right.
4. It shifts the signal higher or lower to correct for drift or to fine-tune control.
5. It adjusts the signal to match real-world values, ensuring the system behaves as expected.

Answers

Worksheet 9 – Temperature Sensor

Questions:

1. What does a temperature sensor do in a PLC system?
2. What type of temperature sensor is used in this worksheet?
3. Why is a signal converter needed with an RTD sensor?
4. What does the PLC do with the analogue voltage from the sensor?
5. What might a raw value near 32,000 mean?

Answers:

1. It tracks heat levels to help prevent overheating, freezing, or system faults.
2. A PT100 RTD sensor.
3. Because the PLC can't read resistance directly—it needs a 0–10 V voltage signal.
4. It converts the voltage into a raw value, then multiplies it by a scalar to get temperature in °C.
5. It usually means a fault like a disconnected sensor, wrong input type, or wiring issue.

Answers

Worksheet 10 – Digital Outputs

Questions:

1. What are digital outputs used for in a PLC system?
2. What happens when you press and hold input I0?
3. How does input I1 behave differently from I0?
4. What does the selector switch (I4/I5) control in this worksheet?
5. Why is it useful to recognise output patterns like the LED sequence?

Answers:

1. To control external devices like lights, motors, relays, and solenoids.
2. LED Q0 turns on only while the button is pressed.
3. LED Q1 stays on after the button is released until turned off by I2.
4. It changes the behaviour of LED Q3. Left keeps it on, right makes it flash.
5. Because real systems often use sequences and recognising them helps with fault finding and confirming correct operation.

Answers

Worksheet 11 – Motor PWM

Questions:

1. What does PWM stand for, and what is it used for in PLC systems?
2. How does PWM control the speed of a motor?
3. What role does the potentiometer play in this worksheet?
4. What does a 100% duty cycle mean compared to 0%?
5. Why is it important to understand PWM when working with motors?

Answers:

1. PWM stands for Pulse Width Modulation, and it is used to control motor speed.
2. It changes how long the output is on versus off, controlling how much power the motor receives.
3. It sets the PWM level by sending a 0-10 V signal to the PLC.
4. 100% means the motor is fully on, and 0% means it is off.
5. Because many real systems use PWM to control speed, and being able to read and adjust it helps with troubleshooting and setup.

Answers

Worksheet 12 – Motor Relay

Questions:

1. What is the main purpose of a relay in a PLC system?
2. How does the PLC activate a relay?
3. What happens when the relay is energised?
4. Why can't PLCs always control devices like motors directly?
5. What should you check if a relay does not switch as expected?

Answers:

1. To safely control higher-power devices like motors or solenoids.
2. By sending power to a coil inside the relay.
3. The relay closes internal contacts to switch the connected output on.
4. Because PLC outputs are low-power and need a relay to handle larger loads.
5. Check for stuck contacts, faulty coils, incorrect logic, or wiring issues.

Answers

Worksheet 13 – Lockout Tagout

Questions:

1. What is the purpose of Lockout Tagout (LOTO)?
2. Why is pressing an emergency stop not enough for safe maintenance?
3. What does installing a padlock on a LOTO device prevent?
4. Can the PLC still receive commands while LOTO is applied? What happens?
5. Who should keep the key to the LOTO padlock?

Answers:

1. To ensure equipment is properly isolated and cannot be restarted during maintenance.
2. Because someone could reset it or restart the system, creating a safety risk.
3. It prevents the emergency stop from being reset.
4. Yes. The PLC receives commands, but the system remains inhibited and will not run.
5. The person carrying out the maintenance should keep the key.

Version Control

VERSION CONTROL

11 07 25	First Revision Created
25 03 26	Added LOTO worksheet